Unit 1 - Sem 4 - 22MAT230

Mathematics for Computing 4

Dr Sunil Kumar S and Prof K P Soman

School of Artificial Intelligence

Amrita Vishwa Vidyapeetham

If you find any mistakes or have any comments to share,

I would be grateful to receive them at s_sunilkumar@cb.amrita.edu

https://github.com/mfcpjt/MFC4_22MAT230

```
clearvars
clear all
ready = true;
PUBLISH = ready;
```

```
clearvars -except PUBLISH ready
```

BiPIL

MNIST digits data set classification

The 28×28 images are flattened and kept as columns in the data matrix X.

The weights $W_{1,2}$ are initialized with $U(-1, 1)$.

Step 1

Define a nonlinear invertible $\phi(x)$

$$W_{1,2} \sim U(-1, 1)$$

$$H_1 = \phi(W_{r1}X)$$

$$H_2 = \phi(W_{r2}H_1)$$

$$W_o = TH_2^+ \qquad \because \; W_oH_2 = T$$

In step 1, only W_o is infused with information regarding data and labels. W1 and W2 remain random.

Step 2

$$H_{2b} = W_o^+T$$

$$H_{1b} = W_{r2}^+\phi^{-1}(H_{2b})$$

Step 3

$$W_{1f} = \phi^{-1}(H_{1b})X^{+}$$

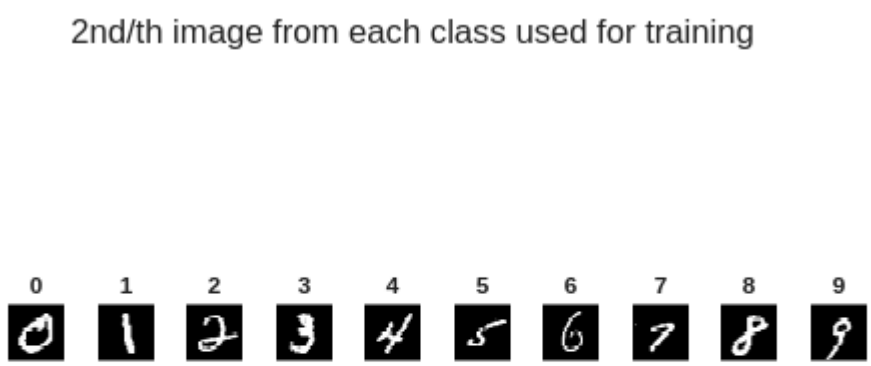
$$H_1 = \phi(W_{1f}X)$$

$$W_{2f} = \phi^{-1}(H_2)H_1^{+}$$

```
clearvars -except PUBLISH ready
tic
load("/media/user/DATA4LINUX/new1/Repos/Mine/MFC4_22MAT230_prep/
MNIST_XY100.mat")
```

```
N = 100; % No: data from each class
k = 0:9;  % classes
n = length(k); % no classes
d = size(X,1); % 'raw' features 28 x 28
true_labels = kron((1:10)',ones(N,1));
```

```
sample = 2;
for i = 1:n
    subplot(1,10,i)
    imshow(reshape(X(:,(i-1)*100+sample),[28,28]))
    title(num2str(i-1))
end
sgtitle(num2str(sample)+"nd/th image from each class used for training")
```



```
nn = [2000, 100];  % # neurons in 1st layer & 2nd layer
```


Anonymous function Format `handle = @(inputs) expression;`

`Example sqr = @(x) x.^2;`

`result = sqr(5);`

```
phi = @(z) cos(z);
phi_inv = @(z) acos(z);
```

Step 1

Define a nonlinear invertible function $\phi(x)$

$$W_{1,2} \sim U(-1, 1)$$

$$H_1 = \phi(W_{r1}X)$$

$$H_2 = \phi(W_{r2}H_1)$$

$$W_o = TH_2^+ \qquad \because\ W_oH_2 = T$$

```
W1 = 2*rand(nn(1),d)-1;
W2 = 2*rand(nn(2),nn(1))-1;

H1 = phi(W1*X);
H2 = phi(W2*H1);
Wo = Y*pinv(H2);
```

Step 2

$$H_{2b} = W_o^+T$$

$$H_{1b} = W_{r2}^+\phi^{-1}(H_{2b})$$

```
H2 = pinv(Wo)*Y;
H1 = pinv(W2)*phi_inv(H2);
```

Step 3

$$W_{1f} = \phi^{-1}(H_{1b})X^{+}$$

$$H_1 = \phi(W_{1f}X)$$

$$W_{2f} = \phi^{-1}(H_2)H_1^{+}$$

```
W1b = phi_inv(H1)*pinv(X);
H1b = phi(W1b*X);
W2b = phi_inv(H2)*pinv(H1b);
H2b = phi(W2b*H1b);
```

```
%% Perdict forward=====================================================
pred_scores = Wo*phi(W2*phi(W1*X));
[~, pred_labels] = max(pred_scores);
% size(pred_labels)
% size(Y)
train_accuracy = mean(pred_labels' == true_labels);
fprintf('Training Accuracy using Forward Weights: %.2f%%\n', train_accuracy
* 100);
```

```
Training Accuracy using Forward Weights: 35.80%
```

```
%% Perdict backward====================================================
pred_scores = Wo*phi(W2b*phi(W1b*X));
[~, pred_labels] = max(pred_scores);
train_accuracy = mean(pred_labels' == true_labels);
fprintf('Training Accuracy using backward Weights: %.2f%%\n', train_accuracy
* 100);
```

```
Training Accuracy using backward Weights: 100.00%
```

```
HH = [H2; H2b];
WHH = Y * HH';
pred_scores = WHH * HH;   % k x N
%% Perdict using both backward and forward
[~, pred_labels] = max(pred_scores, [], 1);  % 1 x N vector
train_accuracy = mean(pred_labels' == true_labels);
fprintf('Training Accuracy using both weights: %.2f%%\n', train_accuracy *
100);
```

```
Training Accuracy using both weights: 100.00%
```

```
toc
```

```
Elapsed time is 2.191622 seconds.
```

Reference

School of AI, AVV

https://ieeexplore.ieee.org/document/11005469

Journals & Magazines > IEEE Transactions on Neural N... > Volume: 36 Issue: 9

Bi-PIL: Bidirectional Gradient-Free Learning Scheme for Multilayer Neural Networks

Publisher: IEEE Cite This PDF

Ke Wang ; Binghong Liu ; Pandi Liu ; Yungao Shi ; Ping Guo ; Yafei Li **All Authors**

```
if(PUBLISH == ready)
    path = '/media/user/DATA4LINUX/new1/Repos/Mine/MFC4_22MAT230/';
    mlxfile = matlab.desktop.editor.getActive().Filename;
    [~, name, ext] = fileparts(mlxfile);
    outfile = [path, name, ext,'.pdf']
    export(matlab.desktop.editor.getActive().Filename, outfile);
    if ispc
        winopen(outfile);
    elseif ismac
        system(['open ' char(outfile)]);
    else
        system("env -u LD_LIBRARY_PATH xdg-open '" + outfile + "' &");
    end
end
```

```
outfile =
'/media/user/DATA4LINUX/new1/Repos/Mine/MFC4_22MAT230/MISC_BiPIL.mlx.pdf'
```